

ABHINAV PARAMESHWARAN

AI Engineer | Machine Learning & Data Science | LLM Systems & Agentic AI
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EDUCATION

University of Southern California

M.S. in Computer Science

Dec 2024

University of Mumbai

B.E. in Computer Engineering

May 2022

EXPERIENCE

Keck School of Medicine of USC

AI Data Scientist

Los Angeles, CA
Nov 2023 – Present

- Designed and deployed automated data processing pipelines handling **10TB+** of **scRNA-seq** and **PacBio** sequencing data, supporting **computational biology** research across 3+ active studies and reducing analysis turnaround time by **~40%**.
- Built a **classification model** for **anomaly detection** in genomic data achieving **92%** accuracy; automated erroneous sample flagging to reduce manual QC overhead by **10%**.
- Developed a custom **statistical QC algorithm** for **PacBio** sequencing validation using sex-marker expression as ground truth, reducing false-positive variants by **99%** and saving thousands in downstream lab validation costs.
- Built **LLM-based** research assistant tooling using domain-specific **prompt engineering** and configured **Claude** workflows with computational biology context to accelerate researcher dry-lab tasks (literature review, code generation, **multi-omics** interpretation).
- Implemented multi-sample **scRNA-seq integration** and **normalization pipelines**; standardized protocols to eliminate batch effects across pooled donor samples and streamline downstream analysis.

Accenture

Software Developer

India
Jun 2022 – Apr 2023

- Developed **Python** automation utilities and **REST APIs** using **Flask**; integrated **SQL/Oracle** databases, reducing processing time by **25%**.
- Wrote **pytest** unit tests and resolved **50+** production bugs; contributed to **CI/CD** pipelines for enterprise clients.
- Supported **5+** enterprise client projects in **Agile** environment; delivered **20+** technical documents and requirement specifications.
- Refactored legacy **Python** modules and optimized **SQL** database queries, reducing **API** response latency by **~150ms** for client portal services.

TECHNICAL SKILLS

AI Engineering: LLMs, RAG, Prompt Engineering, LangGraph, LangChain, MCP, Semantic Search, Pinecone, Claude API

Machine Learning: Classification Models, Anomaly Detection, Deep Learning, NLP, Model Evaluation, scikit-learn

Bioinformatics: scRNA-seq, PacBio, Multi-omics, Single-Cell QC Pipelines, Statistical Modeling

Data & Backend: Python, SQL/Oracle, Pandas, NumPy, R, ETL Pipelines, REST APIs, Flask, pytest

Cloud & Tools: AWS, Docker, Git, CI/CD

PROJECTS

Multi-Agent Trip Concierge & Logic Supervisor | *Python, LangGraph, Pinecone, REST API*

Feb 2026

- Built a supervised **multi-agent** system using **LangGraph** conditional routing and **Pinecone RAG** to ingest unstructured chat logs and resolve scheduling conflicts under venue constraints.
- Implemented **supervisor pattern** with **state-aware routing** across specialized agents; benchmarked conflict resolution across multiple venue-type combinations.

AI Teaching Assistant | *MCP, Pinecone, React, RAG*

Dec 2025

- Built a multi-agent **RAG** chatbot using **Pinecone** and **MCP** to support course-content Q&A through a **React** frontend serving **300+** students.
- Integrated **MCP** for dynamic course material ingestion; implemented response grading and user feedback loops for continuous **model refinement**.

Social Media Engagement Prediction (DUNEs of Influence) | *Deep Learning, Python, NLP*

Jun 2025

- Designed a **deep learning** model to predict social media engagement trained on a custom dataset of **50K+** posts, outperforming baseline models by **25%**.
- Experimented with **transformer-based** embeddings and temporal features; achieved **0.78 RMSE** on hold-out test set; deployed **inference API** for real-time predictions.